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Syllabus

MATH 574 Bayesian Computational Statistics

Instructor

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Course Description

Rigorous introduction to the theory of Bayesian Statistical Inference and Data Analysis, including prior and posterior distributions, Bayesian estimation and testing, Bayesian computation theories and methods, and implementation of Bayesian computation methods using popular statistical software.

Course Outcomes

Upon successful completion of this course, you will be able to:

- Students will learn fundamental theories of Bayesian statistics,
- Students will learn Bayesian computational theories and techniques for modeling and drawing inferences from data sets,
- Students will learn to use visual and numerical diagnostics to assess the soundness of Bayesian modeling,
- Students will become familiar with the computational requirements and compromises to be made in analyzing data sets, and
- Students will gain experience in analyzing real data sets and communicating their results.
- Students will learn how to implement and use these numerical methods in R (or another similar software package),
- Students will improve their presentation and writing skills.

Course Materials

The link to reading materials and resources to learn on the topics can be found in each week's learning module. All materials are available online for free, no required resources need to be purchased. There is no required textbook to supplement the course materials. Note: Be aware that some resources may open in a new tab.

Software Requirements: R or Python

Textbook:
Gelman, A., Carlin, J. B., Stern, H. S., Rubin, D. B. (2013) Bayesian Data Analysis, Third Edition, Chapman & Hall/CRC.

Course Outline

The course consists of 8 modules that focus on the following key areas:

Module 1: Fundamentals of Bayesian Inference

Key concepts

- Bayesian Probability
- Bayesian Inference
- Computation

Readings

- What is Bayes' theorem, and how can it be used to assign probabilities to questions such as the existence of God? What scientific value does it have?
- Simple Bayesian Problems: The Sally Clark Case.
- The Prosecutor's Fallacy
- Chapter 1: Probability and inference of BDA3
- DownloadingR_Directions.pdf. A document explaining how to download R onto your system.
- DownloadingRStudio.pdf. A document explaining how to download RStudio, an editor for R, onto your system.

Module 2: Single Parameter Models

Key concepts

- Estimating Probabilities and Posterior Distributions
- Summarizing Posterior Inference and Prior Distributions
- Normal Distribution and Other Single-Parameter Models

Readings

- Chapter 2: Single-Parameter Models of BDA3

Module 3: Multiparameter Models

Key concepts

- Handling Normal Data and Nuisance Parameters
- Conjugate Priors and Multivariate Normal Models
- Advanced Multivariate Models and Practical Applications

Readings

- Chapter 3: Introduction to Multiparameter Models of BDA3
- Bayesian data analysis demo 3.6

Module 4: Large-Sample Inference and Frequency Properties

Key concepts

- Normal Approximation and Its Applications
- Exploring Large-Sample Theory and Counterexamples
- Frequency Properties and Broader Interpretations of Bayesian Methods

Readings

- Bootstrap Methods: Another Look at the Jackknife (1979) by B. Efron
- The Bayesian Bootstrap (1981) by D. B. Rubin
- Chapter 4: Asymptotics and connections to non-Bayesian approaches of BDA3

Module 5: Hierarchical Models

Key concepts

- Parameterized Priors and the Concept of Exchangeability
- Analysis and Applications of Hierarchical Models
- Computational Techniques and Model Validation

Readings

- Chapter 5: Hierarchical Models in BDA3

Module 6: Bayesian Computation

Key concepts

- Numerical Methods and Approximations in Bayesian Computation
- Simulation Techniques for Bayesian Inference
- Advanced Markov Chain Methods for Bayesian Computation

Readings

- Chapter 10: Introduction to Bayesian computation in BDA3.
- Chapter 11: Basics of Markov chain simulation in BDA3. This chapter in the textbook discusses Markov chain simulation, Gibbs sampler, Metropolis and Metropolis-Hastings algorithms.
- First Links in the Markov Chain
- Markov Chain Simulator

Module 7: Regression Models

Key concepts

- Foundations of Bayesian Regression Analysis
- Advanced Techniques in Hierarchical Linear Models
- Exploring Generalized Linear Models in Bayesian Context

Readings

- Chapter 14: Introduction to Regression Models in BDA3
- Chapter 15: Hierarchical Linear Models in BDA3
- Chapter 16: Generalized Linear Models in BDA3

Module 8: Mixture Models

Key concepts

- Setting Up and Interpreting Mixture Models
- Practical Applications and Computational Challenges

Readings

- Chapter 22: Finite Mixture Models in BDA3

Course Structure and Learning Activities

There are 8 content modules in this course and each module may take about 9-12 hours to complete. The final module consists of your final exam for the course.

This course is comprised of the following elements:

Learning Activities:

- Readings: Each module may include several required and/or supplemental readings.
- Video Lessons: In each module, the concepts you need to know will be presented through a collection of short videos. You may stream these videos for playback within the browser by clicking on their titles.
- Practice Quizzes: Each module will include some practice quizzes, intended for you to assess your understanding of the topics. You will be allowed unlimited attempts at each practice quiz. There is no time limit on how long you take to complete each attempt at the quiz. These quizzes do not contribute toward your final score in the class.

Assessments:

- Discussion Forum: This course has a place for you to interact with other learners about class-related topics.
- Summative Module Assessments: Each module will include at least one summative module assessment. You will be allowed one attempt every eight hours for each assessment. There is no time limit on how long you take to complete each attempt at the assessment. Your highest grade will be recorded.
- Final Assessment: This course will contain one summative course assessment. You will be allowed one attempt for the assessment. Before taking the exam, please make sure you are in a place with reliable internet connection. No retakes will be granted for the lack of internet access. You are in an online program and the use of the Internet is a requirement.

How to Pass This Course

Guidelines for completing and submitting each assigned course activity is posted along with the assignment. Assignments can be submitted at any time as you move through the module. *No late assignments will be accepted.* In case of extenuating circumstances beyond your control that prevent the submission of an assignment or exam, you have to enter a request with the program advisor and the instructor.

To qualify for a Course Certificate, simply start verifying your coursework at the beginning of the course and pay the fee. Coursera Financial Aid is available to offset the registration cost for learners with demonstrated economic needs. If you have questions about Course Certificates, please see the help topics here.

Also note that this course is part of the Master of Data Science program offered by Illinois Institute of Technology. By earning a Course Certificate in this course, you are on your way toward earning a Specialization Certificate in this topic. See more information about the program here.

If you choose not to pay the fee, you can still audit the course. You will still be able to view all videos, submit practice quizzes, and view required assessments. Auditing does not include the option to submit required assessments. As such, you will not be able to earn a grade or a Course Certificate.

The following table explains the breakdown for what is required in order to pass the class and qualify for a Course Certificate. You must pass each and every required activity in order to pass this course.

Activity	Required?	Number per Course	Estimated Time per Module	% Required to Pass	% of Total Grade
Lecture Videos	Yes	3-6 per module	.5-1 hour	N/A	N/A
Practice Quizzes	No	3-6 per module	.5 hour	N/A	N/A
Discussions	Yes	1 per course	1 hour	N/A	N/A
Summative Module Assessments	Yes	1 per module	.5 hour	80%	7.5%/each module (60%)
Summative Course Assessment	Yes	1-2 per course	1-3 hours	80%	40%

Letter Grades

Letter grades are used for the final grade. Information about IIT grading system can be found in the Graduate Student Handbook.

Letter Grade	Description	Points	Percentage
A	Excellent	4.00	90-100
B	Above Average	3.00	80-89.99
C	Average	2.00	70-79.99
E	Fail	0.00	Under 70%

Getting and Giving Help

- Use the Learner Help Center to find information regarding specific technical problems. For example, technical problems would include error messages, difficulty submitting assignments, or problems with video playback. If you cannot find an answer in the documentation, you can also report your problem to the Coursera staff by clicking on the *Contact Us!* link available on each topic's page within the Learner Help Center.
- Use the flag icon under each item to report errors in lecture video content, assignment questions and answers, assignment grading, text and links on course pages, or the content of other course materials.
- Familiarize yourself with Coursera's policy on Accessibility.

Academic Integrity

Your attentiveness to academic integrity reflects the value you place on your own work and the work of others. In addition to Coursera's Honor Code, we also have high expectations for conduct during course participation.

Discussion Forums: Expectations

Sharing an online course with other avid learners like you gives you a unique opportunity to share, collaborate, and learn from others and their experiences, and helps you reinforce your understanding of the topics of the course. Interacting in the Discussion Forums is a great way to engage with your online community. We know that it is not possible to read every discussion forum post, so we recommend that you read those that interest you; and reply when you can contribute. The forum is part of your class activities and everybody is expected to act professionally and be civil and respectful of others in your class. Failure to meet these expectations may be considered a break in the Academic Code of Conduct and may result in your removal from the course. Please, check tips and helpful tools to interact in discussion forums in this document.

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